

2025 AMC 10A Problem 12 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 12. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 12

Problem:

Carlos uses a 4-digit passcode to unlock his computer. In his passcode, exactly one digit is even, exactly one (possibly different) digit is prime, and no digit is 0. How many 4-digit passcodes satisfy these conditions?

Answer Choices:

- A. 176
- B. 192
- C. 432
- D. 464
- E. 608

Solution:

Note that 2 is both even and prime. As we are required exactly one of each, we take cases:

Case 1: The digit 2 is used (exactly once).

Then no other even digits (4, 6, 8) and no other primes (3, 5, 7) may appear. The remaining three positions must be filled with 1 or 9 only. The possible multisets and their permutation counts are:

Multiset	# codes
$\{2, 1, 1, 1\}$	$\frac{4!}{3!} = 4$
$\{2, 1, 1, 9\}$	$\frac{4!}{2!} = 12$
$\{2, 1, 9, 9\}$	$\frac{4!}{2!} = 12$
$\{2, 9, 9, 9\}$	$\frac{4!}{3!} = 4$

This gives us a total of $4 + 12 + 12 + 4 = 32$.

Case 2: Use one odd prime $p \in \{3, 5, 7\}$ and one even $e \in \{4, 6, 8\}$, each exactly once.

There are $\binom{3}{1} \cdot \binom{3}{1} = 9$ choices of (p, e) . The other two positions must be from $\{1, 9\}$. For a fixed (p, e) , the possible multisets and their counts are:

Multiset	# codes
$\{p, e, 1, 1\}$	$\frac{4!}{2!} = 12$
$\{p, e, 1, 9\}$	$\frac{4!}{1!} = 24$
$\{p, e, 9, 9\}$	$\frac{4!}{2!} = 12$

This gives us a total of $12 + 24 + 12 = 48$ per each (p, e) , or an overall total for the case of $9 \times 48 = 432$.

Therefore, our total number of passcodes is

$$32 + 432 = \boxed{\text{(D) } 464}.$$

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